

PERITONEAL CAVITY & RETROPERITONEUM — Imaging Revision Table

Bold = high-yield. CT = mainstay; US/MRI = no ionizing radiation. Radiology lecture (Dr Kawa Mahmood).

PERITONEAL CAVITY

Condition	Cause	Imaging findings
Ascites	Small amounts detected US/CT/MRI; nature NOT reliably differentiated	US detects smallest amount. CT = low density. Free → dependent sites: pouch of Douglas, Morrison's pouch, paracolic gutters. Loculated → mimics abscess
Peritoneal tumours	Mostly metastases — ovarian most important	(a) ascites only, (b) peritoneal nodules , (c) omental cake
Intra-peritoneal abscess	Follows perforation ; site depends on perforation; multiple common; ascites frequent	US : loculated + irregular wall, internal echoes, air; DDx bowel (peristalsis). CT : fluid + enhancing wall, air in 50% (bubble/streak/air-fluid level). In-111/Tc leukocyte scan if US/CT fail (e.g. anastomotic leak)

RETROPERITONEUM — ANATOMY & MASSES

Anatomy / modalities	3 compartments (anterior + posterior renal fascia). Organs: adrenal, aorta, retrocrural, IVC, psoas. CT main ; plain film (mass/calcification/gas); MRI (multiplanar, vascular extension)
Mass DDx (7)	1. Nodal (lymphoma, germ cell, mets) 2. Retroperitoneal fibrosis 3. Neurogenic (paraganglioma, schwannoma, ganglioneuroma) 4. Sarcoma (lipo/fibro) 5. Haematoma (trauma/ruptured AAA) 6. Infection (psoas abscess) 7. Vascular (aneurysm)
Lymphadenopathy	Mets & lymphoma look identical → FDG-PET/CT (esp lymphoma follow-up). Normal short axis: para-aortic 10mm, retrocrural 6mm; >20mm = invariably malignant. Lymphoma → confluent lobulated mass around aorta
Retroperitoneal tumours	Mainly liposarcoma/fibrosarcoma; nature usually undeterminable. Liposarcoma recognizable = fat + strands of soft tissue density

ADRENAL IMAGING (CT best routine)

Non-functioning adenoma	<3cm hard to tell from mets; >3cm = mostly mets (rarely carcinoma). Metastases common, frequently bilateral, lung common primary
Cushing's adenoma	Always >2cm, always seen CT/MRI; <10 HU density
Aldosteronoma	Usually <1cm — difficult to identify
Hyperplasia	Normal / mild enlargement
Phaeochromocytoma	Very large, all detected; 10% rule (multiple, bilateral, extra-adrenal, malignant); MIBG scan for localization

AORTIC ANEURYSM (AAA)

Screening / dimensions	US = screening ; normal 2.5cm, >3cm = aneurysm. CT + US = true dimensions. Plain film lateral = calcification. Angiography ≠ true size (good for branch stenosis)
Findings / rupture	CT: blood clot lining wall, calcification, IVC displacement. >6cm = high rupture risk. Leaking/bleeding aneurysm easily diagnosed on CT (haemorrhage in retroperitoneum)